

can be quite irritating. Another point of concern is the absence of the shortcut panel on the status bar that provided quick access to many functions such as Show Movie Explorer, Show Actions, etc.

Design features and elements

Macromedia has introduced some outstanding features that increase productivity considerably. Flash MX incorporates an all-new Transformation tool featuring Distort and Envelope modes. The Envelope mode makes it possible to warp and edit vector objects. The Distort mode allows you to distort the object by its edges or corners and taper it to give it a 3D look. One glitch with both these modes is that you cannot modify symbols, bitmaps, video objects, sounds, gradients, object groups and text. If a multiple object selection contains any of these, only the shape objects are distorted. To modify text, you will have to first convert the characters to shape objects.

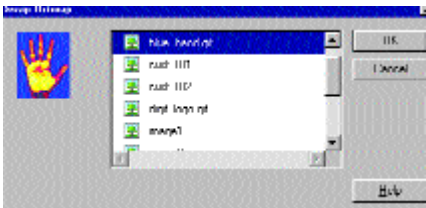
Flash MX also provides a sorely missed element—pixel perfect alignment of objects. You can turn on pixel snapping using the Snap to Pixels command in the View menu. A pixel grid will appear when the view magnification is set to 400 per cent or higher. The pixel grid represents the individual pixels of your frame and when you create or move an object, it is constrained to the pixel grid.

A long-time complaint of Flash users has been the absence of rich media support, i.e. video and audio. Flash MX comes with the Sorenson Spark codec, which means you can import video into Flash in different file formats (AVI, MPEG, etc). When you export the SWF file, the video will be embedded into your Flash movie.

Loading external files is not restricted

to just MP3s. JPEG images can also be imported at run time, allowing you to program events within a Flash movie. This will significantly reduce the overall size of the Flash movie and make the content more dynamic. Another addition is the Font Mapping feature—if you work with a document containing fonts that aren't installed on your system, they will be substituted by those already installed. Alternatively, you could set a default font under *Edit > Preferences > General > Font Mapping Default*.

You can now update and swap symbols in any Macromedia Flash MX document on your computer or on the network. Hence, shared symbols across multiple documents can be manually or automatically updated when a change is made to the original. In case you want to swap images, you don't have to go through the tedious procedure of re-importing the new image. Simply right-click on it and select Swap Bitmap, which will open a panel from which you can select the correct image.



Swap Bitmap allows you to enlarge images without having to reimport the image

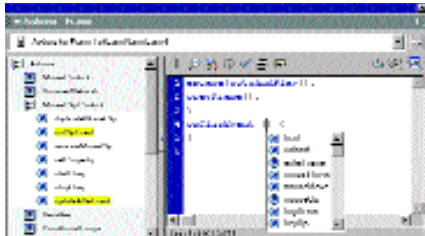
Flash MX also introduces dynamic masking support. Previously, masking could only be done through layers. Now it is possible through ActionScripts that allow you to mask one movie clip object over another, irrespective of the layer it is on. Lastly, the option of allowing the user to save his work as a template gives the flexibility to reuse the saved document as a starting point for other Flash documents.

Development and server-side features

Those who used Flash 4 were overwhelmed by the huge set of ActionScripts, and the inclusion of XML socket commands.

JavaScript support in Flash 5 was a welcome gesture. But Macromedia wants to position Flash MX as a development platform for building content-rich Web-based applications. Hence, JavaScript support has been improved while the ActionScript editor and debugger sport several enhancements.

The ActionScript editor now not only highlights keywords and values in the ActionScript code, but can also detect the

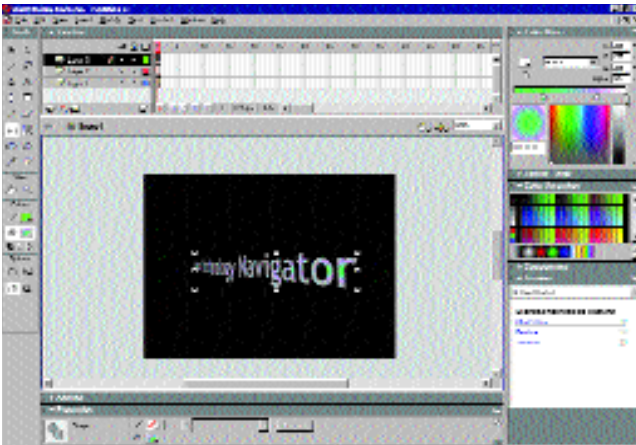


Code Hints make the process of writing scripts a whole lot easier

action you are entering and display a code hint—a tooltip containing the complete syntax for that action or a popup menu listing possible property names and methods. The editor is now a collapsible panel in the interface (owing to the MX panel management system), so you don't need to open a new window every time you want to edit a script. The Script Debugger has also been improved and lets you set breakpoints in your code and step through your scripts one line at a time as your animation runs in Preview mode. This considerably reduces the time taken to debug errors.

If you are still not impressed, then check out the all-new Flash UI components tool—it allows you to quickly and easily add simple user interface elements such as radio boxes, check boxes and drop down lists to your Flash document. As with Visual Interdev, once you have the element on the page, you can modify its parameters. These UI components can also be controlled programmatically with ActionScripts, which in turn can communicate with the server-side programming language.

In the previous version, the code generated by using ActionScripts was different and more complicated. To receive data from a server you had to use the 'LoadVariablesNum' action. However, in Flash MX it has been replaced with just 'load'. This will reduce the amount of coding time put in by expert coders who like to manually enter their code.



The Distort and Envelope tools allow you to warp and edit vector objects